

The University of Lahore
Department of Computer Science& IT
Final-term Examination



Spring-2023

Course Title:	Discrete Structures	Course Code:	CS-09202	Credit Hours:	3
Course Instructors:	Mr. Tayyab, Miss Azka, Miss Samra, Miss Tehseen	Program Name:	BSCS		
Time Allowed:	120 Minutes	Maximum Marks:	45		
Date:	08-06-2023	Course Mentor: Dr. Mudasser Naseer Signature:			
Student’s Name:					
Reg. No:		Section:			

Important Instructions / Guidelines:

- Understanding of the questions is the part of the question paper so does not disturb the invigilators.
- Kindly leave your mobile phones outside exam hall.
- Solve all questions on the answer sheet provided separately.
- Your paper will be marked as ZERO, if involved in any kind of cheating.

Question # 1 Let $A = \{1,2,3,4\}$ and f and g are functions from A to A such that

$$f(1) = 1, f(2) = 4, f(3) = 1, f(4) = 4 \text{ and } g(1) = 2, g(2) = 3, g(3) = 4, g(4) = 1$$

- For each of f and g , determine if it is one-to-one, onto and bijective. [5]
- Check if f and g are invertible functions if any of them is invertible find its inverse function. [5]
- Give the composite functions $f \circ g$ and $g \circ f$. [5]

Question # 2 Prove each of the following by applying the asked method:

- Prove by contraposition that for an integer n , if n^2 is odd then n is odd. [5]
- Apply mathematical induction to show: $2+4+6+\dots+2n = n(n+1)$ for $n \geq 1$ [5]

Question # 3 Use rules of inference to show: [8]

If $\forall x (P(x) \vee Q(x))$, $\forall x (\neg Q(x) \vee S(x))$, $\forall x (R(x) \rightarrow \neg S(x))$ and $\exists x \neg P(x)$ are true then $\exists x \neg R(x)$ is true.

Question # 4 Let $F(x)$: x is a female. $P(x)$: x is a parent. $Mother(x,y)$: x is mother of y .

- Translate from English into logical expression using above predicates and quantifiers. [6]
 - There is a parent who is someone's mother.
 - Every female parent is someone's mother.
 - There is a female who is no one's mother.
- Translate the following logical expression into English. [6]
 - $\forall x \forall y (P(x) \rightarrow (F(x) \vee Mother(x,y)))$
 - $\forall x (\neg P(x) \vee F(x))$
 - $\forall x (F(x) \rightarrow \neg P(x))$